MILESTONE 1 – PROJECT AND DATA DESCRIPTION

Project: Predicting Extreme Particulate Pollution Events Using Explainable Machine Learning: A Case Study of Tetovo, North Macedonia

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Group: G2 | Study period: 2019–2024

Scope: PM2.5 is the primary classification outcome; PM10 is a secondary, separate outcome. Input files are exclusively Data(Final).csv and Meteo_Data_ERA5.csv. No machine learning model is trained in this milestone.

Part A. The problem

Field 1. Problem

The intended users are municipal environmental officers, air-quality monitoring staff, and public-health communication teams considering whether to issue an early warning or intensify monitoring for potential extreme particulate pollution. The decision concerns a warning for the forthcoming 24-hour period at the monitored location, not a citywide definitive exposure estimate. A false negative means a severe future period receives no warning and may delay precautionary communication. A false positive means a warning is issued without an extreme event, using staff attention and potentially reducing trust through repeated unnecessary alerts. Because missed severe events may be more harmful, the evaluation will place particular emphasis on recall alongside false-warning burden.

Field 2. Task

The task is supervised binary classification. At timestamp t, the PM2.5 outcome extreme_PM2.5 is 1 when the arithmetic mean of observed PM2.5 concentrations at t+1 through t+24 exceeds the training-period threshold, and 0 otherwise. The secondary outcome extreme_PM10 is defined analogously. An incomplete future 24-hour target window receives a missing label and is excluded from label counts. The engineered targets are not original CSV columns. Inputs proposed for subsequent modelling are observed PM2.5 and PM10 history (lagged one hour and trailing 24-hour averages), temperature in °C, relative humidity, wind speed, eastward and northward wind components, and predictable calendar features. Labels are derived from subsequently observed outcomes, making this supervised classification rather than regression or unsupervised learning.

Field 3. Known when

Prediction is issued at the end of hour t after the corresponding pollution observations are received. Both pollutants at t-1 and rolling means ending at or before t are historical. Hour, month, day, and season can be known in advance. ERA5 temperature, humidity, wind speed and components at t are available in this retrospective dataset, but operational use would require measurements or weather estimates demonstrably available at issuance time; reanalysis is not a live forecast. All future24_* columns and extreme_* labels, any t+1 through t+24 pollution or weather measurements, and Temperature (Kelvin duplicate of temperature_C) are excluded as predictors. wind_direction is excluded initially because Wind-u and Wind-v already encode direction without a discontinuity at 360 degrees. Data availability must be audited before deployment.

Field 4. Success

Primary metric for later model evaluation: average precision (area-style summary of precision–recall performance), because extreme pollution is a minority-class concern. Secondary metric: recall for

actual extreme events; precision and false-positive rate will be tracked to quantify unnecessary warnings. A final alert decision threshold will be selected using development data, not test results. For Milestone 1, the required no-training majority-class accuracy baseline is the count of the more frequent class divided by all labelled examples in the training period. It does not represent a fitted model or an average-precision baseline.

Pollutant	Training labelled	Non-extreme	Extreme	Majority-class accuracy
PM2.5	30,654	27,588 (90.00%)	3,066 (10.00%)	90.00%
PM10	30,653	27,587 (90.00%)	3,066 (10.00%)	90.00%

Part B. The data

Field 5. Origin

The pollution file supplied for this assignment is Data(Final).csv (hourly PM2.5 and PM10 values). Prior study documentation identifies the national air-quality monitoring portal operated by North Macedonia's Ministry of Environment and Physical Planning as the original measurement source. The meteorological file supplied is Meteo_Data_ERA5.csv, described as ERA5-derived meteorological data; ERA5 is produced by ECMWF through the Copernicus Climate Change Service. Files cover 2019–2024 and were prepared for environmental research. Source references for verification: https://air.moepp.gov.mk/ and https://cds.climate.copernicus.eu/. Exact pollutant station metadata, processing history, authorization to redistribute original measurements, relevant version/download references, and licence terms for each derived file cannot be established from CSV contents alone and must be verified before external release. Possible source-specific signal includes roadside/near-source measurements and meteorology derived at grid-cell scale; either can limit generalization.

Field 6. Content

Each pollution row is a timestamped particulate reading; each ERA5 row is one hourly meteorological record. Raw Data(Final).csv has 52,603 rows and 3 columns, while Meteo_Data_ERA5.csv has 52,608 rows and 8 columns. After averaging conflicting repeated pollution timestamps and left-joining pollution to the complete meteorological timeline, there are 52,608 hourly rows, from 2019-01-01 00:00:00 to 2024-12-31 23:00:00. Concentration units are assumed µg/m³ pending source metadata verification.

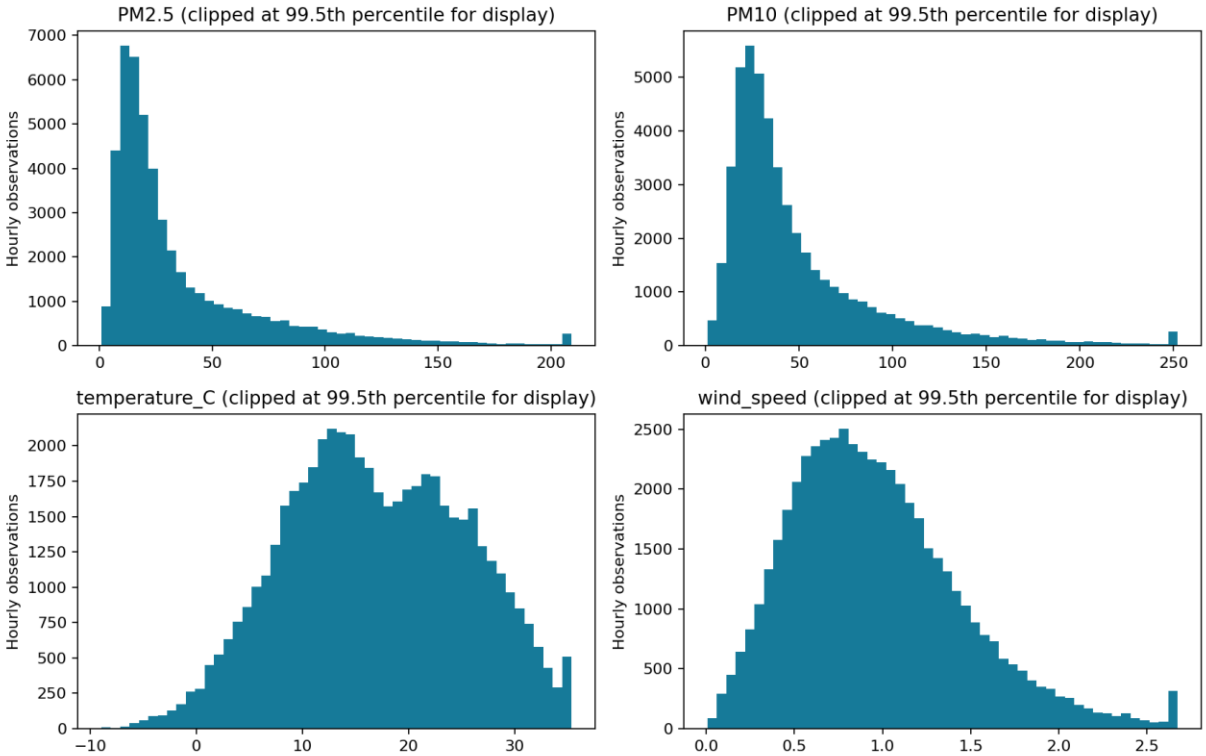
Column	Type	Unit	Meaning
Date	datetime	date/time	Hourly observation time
PM2.5	float	µg/m³ (verify)	Fine particulate matter
PM10	float	µg/m³ (verify)	Coarse-plus-fine particulate matter
temperature_C	float	°C	ERA5 2m temperature in Celsius
Relative Humidity	float	%	ERA5 derived RH; >100% flagged
Wind-u	float	m/s	Eastward wind component
Wind-v	float	m/s	Northward wind component
wind_speed	float	m/s	Derived magnitude
wind_direction	float	degrees	Derived direction
Temperature	float	K	ERA5 temperature in Kelvin, excluded as duplicate

Pollutant / split	Valid targets	Non-extreme	Extreme
PM2.5 / train	30,654	27,588 (90.00%)	3,066 (10.00%)
PM2.5 / validation	8,655	7,981 (92.21%)	674 (7.79%)
PM2.5 / test	8,551	8,051 (94.15%)	500 (5.85%)
PM10 / train	30,653	27,587 (90.00%)	3,066 (10.00%)
PM10 / validation	8,655	8,052 (93.03%)	603 (6.97%)
PM10 / test	8,527	7,983 (93.62%)	544 (6.38%)

Extreme thresholds are fixed using the training-only 90th percentile of future 24-hour means: PM2.5 = 83.888, PM10 = 104.534 in assumed µg/m³. These are dataset-defined severity cutoffs, not regulatory standards. Training = 2019–2022, validation = 2023, final holdout test = 2024.

Field 7. Quality

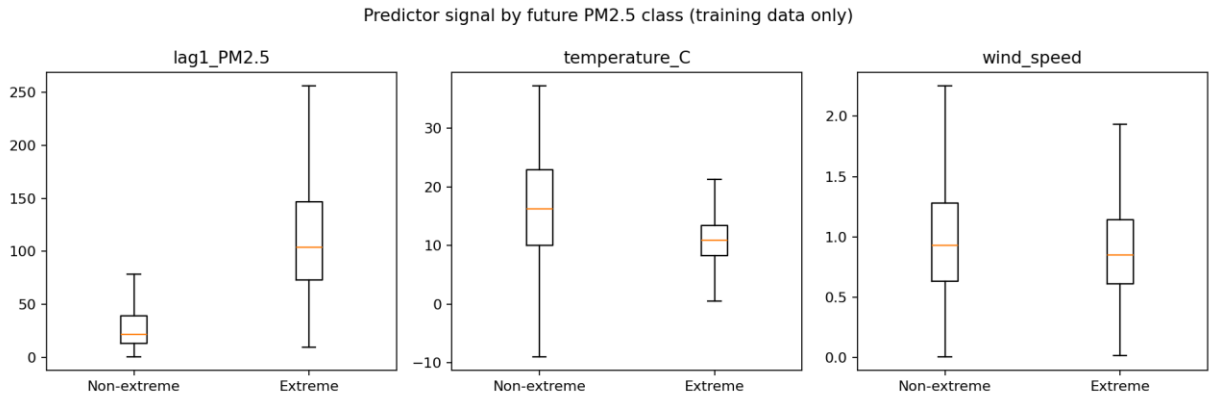
In the original pollution file, PM2.5 has 3,965 missing measurements and PM10 has 3,967; ERA5 has zero empty cells. There are 0 exact duplicate pollution rows and 7 repeated pollution timestamps, leaving 52,596 distinct pollution timestamps. Repeated timestamps contain non-identical readings and were averaged for a reproducible unique-hour merge (not presented as exact copies). The merged hourly timeline contains 3,977 missing PM2.5 and 3,979 missing PM10 cells. These missing pollution inputs will later be imputed within training folds or handled through missingness-aware algorithms; future target windows requiring any unobserved hour are withheld rather than filled to invent outcomes. ERA5 Relative Humidity exceeds 100% in 1,308 rows and requires unit/derivation checks; those readings should be flagged, and any clipping justified and applied consistently. Temperature appears in Kelvin and Celsius; only Celsius will be used to avoid redundant variables. Time-related rows share events and autocorrelation and must be separated chronologically, not randomly.



Field 8. Signal

For the primary PM2.5 label, training-period comparisons contrast predictors observed before the target window for extreme versus non-extreme future 24-hour periods. They are descriptive evidence of candidate signal and do not establish causation, independence, or achievable predictive accuracy.

Available predictor	Non-extreme mean	Extreme mean
lag1_PM2.5	31.340	114.008
temperature_C	16.382	10.592
Relative Humidity	73.061	79.785
wind_speed	0.998	0.880



Historical PM2.5, temperature, and wind-speed distributions differ between future classes, supporting investigation with supervised learning. Temporal autocorrelation means these are not independent samples; only out-of-time testing can establish generalization.

Part C. The risks

Field 9. Risks

Temporal data leakage: Since the project predicts pollution conditions for the following 24 hours, future measurements must not be included as model inputs. To prevent this, only information available at the prediction time will be used, and chronological data splitting will be applied.

Limited geographical coverage: Air-quality measurements come from a single monitoring station in Tetovo, while meteorological data are obtained from ERA5. Therefore, the findings may not represent pollution conditions across the entire city. The conclusions will be limited to the monitored location.

Meteorological data availability: ERA5 provides historical meteorological information rather than real-time weather forecasts. The project will therefore focus on retrospective prediction experiments. Operational use would require meteorological information available at the prediction time.

Missing pollutant measurements: Some PM2.5 and PM10 observations are missing, which may affect data quality. Missing input values will be handled using appropriate

preprocessing methods, while incomplete future observation windows will be excluded from target calculations.

Duplicate timestamps: The dataset contains repeated timestamps with different pollutant measurements. These observations require careful examination and consistent handling to avoid introducing errors into the analysis.

Unusual humidity values: Some relative humidity measurements exceed 100%, indicating potential issues with data processing or calculation. These values will be investigated before deciding how to handle them.

Class imbalance: Extreme pollution events are less frequent than non-extreme conditions. This may cause classification models to favor the majority class. Appropriate evaluation metrics, including recall, precision and average precision, will therefore be used.

Seasonal variation: Air-pollution patterns change across seasons and years. The project will use chronological training, validation and testing periods, with additional evaluation of seasonal performance.

Related hourly observations: Consecutive air-quality measurements are strongly related. Random splitting could therefore produce misleading evaluation results. The dataset will be divided chronologically to reduce this risk.

Data source limitations: The accuracy of the analysis depends on the original monitoring data and meteorological measurements. Station information, measurement units and data-source documentation will be checked to ensure that the dataset is interpreted correctly.

Reproducibility

All numerical results, statistical summaries, and figures presented in this report were generated using the accompanying Jupyter Notebook. The analysis uses the original air-quality and meteorological datasets, with preprocessing and data transformations documented in the code. No predictive models were trained, as Milestone 1 focuses on problem definition and data exploration. The accompanying ZIP contains the datasets, code, and required dependencies to support reproducibility.